Hubbard dynamics on a fixed qubit array: protection, mitigation, and achievable lattice size

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Abstract

We compare three uses of the same nearest-neighbor array: physical-qubit computation with error mitigation, partially protected STAR computation, and fully fault-tolerant surface-code computation. The application is a local doublon observable after a Fermi–Hubbard quench. We organize the comparison around the largest lattice accessible with a given number of physical qubits and one week of execution. A common accounting model separates sampling, mitigation exposure, logical protection, non-Clifford supply, and parallel replicas. Explicit circuit compilation and small-patch dynamics identify the mechanisms that control the comparison: idle exposure depends strongly on the schedule; STAR depends on both total rotation angle and accumulated injection floors; FT depends on supply and consumption together; and Trotter refinement changes the cost of these architectures differently. Current four-step resource curves are conditional circuit comparisons, since that step count fails the observable-accuracy target on both square patches tested. We retain this distinction while developing a common basis for architecture optimization and hybrid strategies. For immediate discussion, we also provide a simpler analytical comparison with experimentally anchored circuit scales and explicit scenario ranges.

1 Back-of-the-envelope+ discussion example

We use simple analytical laws to expose the architectural tradeoffs, with $p = 10^{-3}$, 10 ns gates, 100 ns measurement/reset, 240 ns syndrome cycles, $U/J = 4$, and a one-week budget B. The primary target remains $\epsilon = 0.001$, with 0.01 as a sensitivity. The updated comparison includes NISQ, STAR, FTQC, and the hybrid, and separates assumed effective mitigation from specified PEC references.

Step count informed by the small patches. At $\tau = 1$, the saved 2×2 and 3×3 plaquette calculations give $\delta D\, r^2$ between 0.2178 and 0.2231 for $r = 16, 32, 64$. We round this to $C = 0.22$ and use the discussion prescription

$$\delta D_{\text{model}} = C\frac{\tau^3}{r^2}, \qquad r(\tau,\epsilon) = \left\lceil \sqrt{\frac{C\tau^3}{\epsilon_T}} \right\rceil, \qquad \epsilon_T = \epsilon/4. \tag{1}$$

The full-lattice model does not spend an allowance on spatial truncation. At $\tau = 1$, it therefore uses $r = 10$ for the loose target and $r = 30$ for the strict target. If half this reserve were assigned to spatial truncation, $\epsilon_T = \epsilon/8$ would instead give $r = 14$ and 42. Mixing 10 and 42 would mix these two budget conventions.

Equation (1) is a transferred small-patch scenario, not a large-lattice accuracy calibration. The near agreement of two patches does not prove size independence. Transferring the coefficient from the plaquette ordering to the other implementations, and extending it with τ^3 outside $\tau = 1$, are explicit assumptions. The time power is an editable parameter. The earlier $r = \lceil 4\tau \rceil$ example is archived.

Common sampling and protection. For the effective-mitigation and encoded planning curves we use

$$M_0 = \left\lceil \frac{1.96^2 D_*(1-D_*)}{(\epsilon/2)^2} \right\rceil, \qquad Q_{\text{week}} = q \max\left(1, \left\lceil \frac{M_0 F t_{\text{shot}}}{B} \right\rceil\right), \quad D_* = 0.1. \tag{2}$$

This gives about 1.38×10^6 ideal samples at strict precision. The PEC references use the sufficient Hoeffding count from Eq. (7) instead. Both encoded protection and magic-state quality are budgeted per attempt; for analog-mitigated circuits we retain the reference amplification $\Gamma = \sqrt{F}$. The central delivery model has $b = 4$ effective parallel lanes and $\lceil 1.5(2L^2+6) \rceil + 6b$ data/operation tiles. Distance follows the existing surface-code fit. The lane count is a sustained-delivery assumption, not a compiled dependency schedule.

NISQ and its PEC comparisons. Willow's 6×6, three-step circuit supplies the anchor $G_* = 4372$ entanglers and circuit depth 294 [1]. Our proxies are

$$G_2 = 4372 \frac{r}{3} \left(\frac{L}{6}\right)^3, \qquad D_{\text{circ}} = 294 \frac{r}{3} \frac{L}{6}, \qquad \log F_{\text{eff}} = \log(\eta_*) \frac{p}{10^{-3}} \frac{G_2}{4372}. \tag{3}$$

The central $\eta_* = 100$, with 10–1000 sensitivity, is a chosen effective cost of observable recovery, selection, and training. Residual mitigation bias is assumed affordable; this is not a fit to experimental precision. We also show two PEC references. Two-qubit-only PEC uses the exact inverse channel factor for the rounded gate count, with $\log F \simeq 4pG_2$. A second proxy uses saved full-noise exposure, area, and timing from the compiled $L = 2, 4, 6, 8$ circuits, scaling bulk exposure linearly with r while retaining SPAM separately. Beyond $L = 8$ it extends the same cubic routing proxy. At $L = 4, r = 4$ this reproduces $Q = 2132604$. Its refined-step values are extrapolations, not new compiled circuits. The NISQ band includes both PEC references as well as the assumed effective strategies.

STAR, FTQC, and the hybrid. Termwise STAR has

$$\Theta = (5L^2 - 4L)\tau, \qquad R = [16L(L-1) + 2L^2 - 2]r + 1, \qquad \log F_{\text{STAR}} = 4\alpha p\Theta + 4fR. \tag{4}$$

We use $\alpha = 1.5$, $f = 10^{-5}$, and 18 code clocks per completed rotation divided by b [2, 3]. The scenario range varies f from 0 to 10^{-4} and b from 8 to 1. FTQC uses plaquette counts with logarithmic synthesis and $F = 1$. The hybrid uses the same plaquette workload, injects its variable rotations, and supplies only its fixed T states from the FT source. Its exponent is $4\alpha p\Theta_{\text{dyn}} + 4fR_{\text{dyn}}$.

Both FTQC and hybrid use the rounded cultivation reference: 1024 source qubits, 100 syndrome cycles per state plus $2d$ handoff cycles, and state error 2×10^{-9} [4]. One 15-to-1 cleanup stage is added when needed by $\Gamma N_T q_T \le \epsilon_{\text{magic}}$. Supply time is overlapped with the total analog/T consumption time. We charge nine code clocks per consumed T, optimize 1–16 sources, and vary source service time by 0.5–4. The hybrid need not beat termwise STAR, which already avoids fixed T gates.

For $\tau = 1$, the central areas illustrate the changes:

Lattice	ϵ	r	NISQ effective	STAR	Hybrid	FTQC
4×4	.001	30	1404	71442	73348	5.66×10^6
4×4	.01	10	36	58482	60228	73348
6×6	.001	30	5.5×10^{17}	745890	302520	1.89×10^7
6×6	.01	10	162	124362	103548	131986

At 4×4, changing $r = 4$ to 30 raises the effective NISQ exponent from 1.82 to 13.64, while termwise STAR changes from 0.420 to 0.650 through its floor term. The FT area jumps from 79066 to 5.66×10^6,

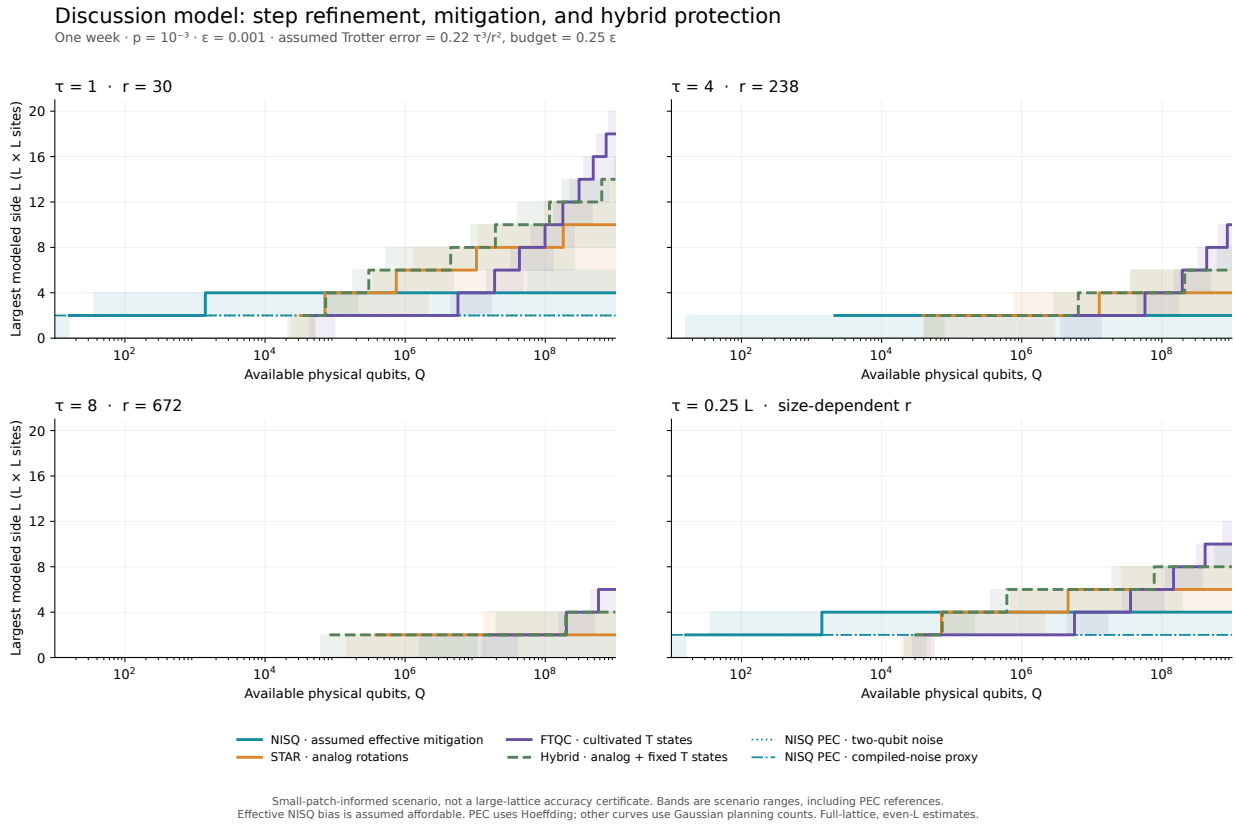


Figure 1: Revised discussion comparison at $\epsilon = 0.001$, using the small-patch-informed step scenario. Lines are central assumptions; bands span scenarios, including the NISQ PEC references, not confidence intervals. The fourth panel follows $\tau = L/4$. All estimates evolve the full lattice and evaluate even L. The assumed effective NISQ line is not a demonstrated accuracy result.

including a cleanup-threshold crossing and extra replicas. Its total cost therefore need not grow merely linearly with r, even though its T count grows approximately linearly apart from synthesis precision.

The time-growing family also depends on the step prescription. With Eq. (1), $\tau \propto L$ gives $r \propto L^{3/2}$ and the chosen NISQ exponent scales as $L^{9/2}$. The previously suggested L^4 instead assumes $r \propto \tau$; it can be explored with time power two in the model error law. For STAR, the angle term scales as L^3 along this family, and the floor term as $L^{7/2}$. FTQC retains polynomial costs with logarithmic protection/synthesis factors. These powers describe the chosen full-lattice implementations; routing, observable support, and lane scaling can change them. The remainder develops the detailed reference models, whose saved numerical artifacts remain separate.

2 The fixed-device question

The same physical qubits can buy more simultaneous samples, stronger logical protection, or faster delivery of non-Clifford operations. We want to determine which use gives the largest accessible dynamics problem, and which combinations can improve on the three reference approaches. Figure 2 shows the resource roles. The comparison includes physical probabilistic error cancellation (PEC) [5], the original and small-angle STAR mechanisms [2, 3], and surface-code computation with explicit magic-state supply [6, 4]. Campbell's Hubbard construction provides a concrete algorithmic baseline [7]; recent partial-FT resource studies [8] and a performed fermionic-dynamics experiment [9] provide comparison

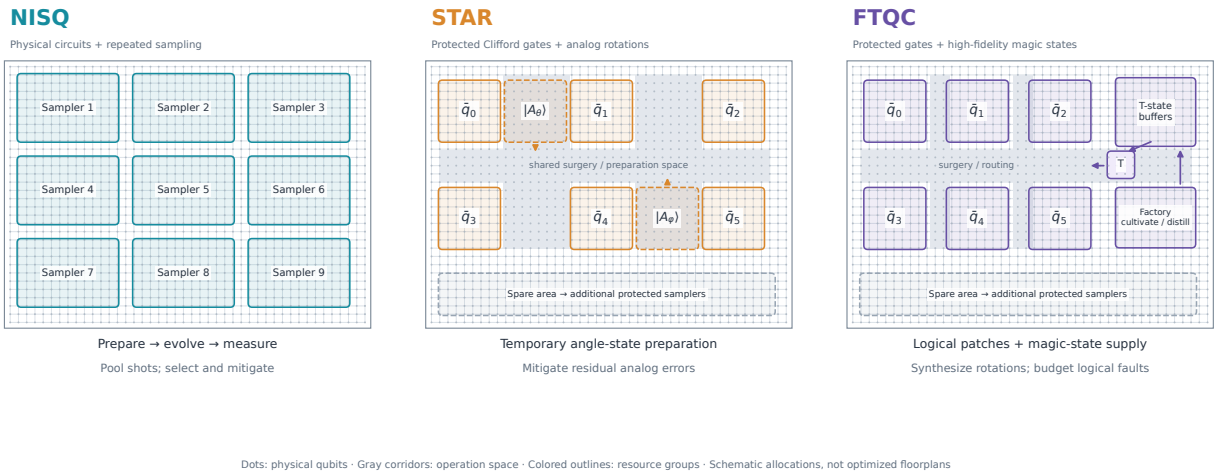


Figure 2: Three allocations of a common sea of physical qubits. NISQ uses many unprotected circuit replicas. STAR and FTQC share protected data patches and lattice surgery. Dashed orange patches show temporary angle-state preparation in STAR surgery ancillas, a possible allocation rather than a dedicated factory requirement [3]. FTQC supplies high-fidelity T states to a synthesized sequence of logical operations. Every dot denotes a physical qubit; each encoded data patch contains many such qubits. The drawings explain resource roles and are not numerical floorplans or equal-capacity implementations of a particular Hubbard lattice.

points with distinct hardware assumptions.

We treat these as specified implementations and estimators. A poor physical PEC curve is not an impossibility theorem for NISQ, and an encoded curve based on reference patch timing is not a validated physical floorplan. The purpose of the analytical laws is to expose the quantities whose measurement or optimization can change the architecture ranking. CUDA-Q Logical supplies explicit circuit and synthesis artifacts [10]; physical encoded timing is kept separate when its implementation or verification is incomplete.

The similarity of the encoded data patches is physical: both approaches protect their Clifford operations. The main distinction is how they supply non-Clifford rotations, including the sequence of operations in time (Figure 5). STAR teleports an angle-specific resource state and mitigates residual rotation errors. Our FTQC reference instead synthesizes the rotation from Clifford gates and high-fidelity T states. In the original STAR preparation protocol, a surgery ancilla can also serve as the preparation patch when distance and physical error rate permit adequate acceptance [3, Sec. II A 2]. Preparation, any patch growth, and retries still occupy space and time. Dedicated injection lanes are therefore a scheduling choice in our cost model, not a defining STAR requirement. Conversely, FTQC supply may be compact or shared in time; neither approach has a universal ratio of data area to preparation area.

3 Task, conventions, and precision

The problem has $N = L^2$ Hubbard sites, $2N$ spin modes, and open boundaries. The physical device has a separate area budget Q. In units where energies are measured in J, we simulate

$$H/J = -\sum_{\langle ij\rangle,\sigma} (c^\dagger_{i\sigma} c_{j\sigma} + \text{h.c.}) + u \sum_i (n_{i\uparrow} - \tfrac{1}{2})(n_{i\downarrow} - \tfrac{1}{2}), \qquad u = U/J = 4. \tag{5}$$

The symmetric onsite form differs from $Un_\uparrow n_\downarrow$ only by a known phase in a fixed-number sector. We prepare the half-filled Néel product state and measure the central doublon $D = \langle n_{i\uparrow} n_{i\downarrow}\rangle$ at $\tau = Jt/\hbar$. Doublon avoids the symmetry-pinned charge-density observable of this quench.

Quantity	Reference value or convention
Connectivity	Square array, nearest neighbors
Gate / measurement / reset	10 / 100 / 100 ns
Physical two-qubit / one-qubit error	$p = 10^{-3}$ / $p/10$
Idle channel	Depolarizing error p per 100 ns
Encoded cycle / classical feedforward	240 / 200 ns, reference assumptions
Runtime cutoff	$B = 604800$ s, one week
Main precision	1% relative at provisional $D_* = 0.1$: $\epsilon = 0.001$
Statistical confidence	$1 - \delta = 0.95$
Rotation convention	$U_P(\theta) = e^{-i\theta P}$; CUDA-Q angle $\phi = 2\theta$

The native physical compilation assumes arbitrary-angle R_{ZZ} and one-qubit pulses at the stated gate duration. The noise channels and encoded cycle are explicit model choices, not a characterization of a fabricated device. In particular, the surface-code suppression fit has not been calibrated to the same physical idle and syndrome schedule.

The target is $\epsilon = \eta|D|$ with $\eta = 0.01$. Resource curves use the declared signal D_*, rather than silently treating it as calculated dynamics. Campbell's energy precision proportional to N corresponds to fixed absolute precision in energy density [7]. Our local D is already intensive: fixed absolute error in D does not itself become stricter with size. Relative precision is useful when the signal magnitude is the relevant scale. Strict relative precision at zero signal needs a separate prescription; a production estimator with unknown nonzero D would require a pilot or sequential confidence procedure. We retain $\epsilon = 0.01$ absolute as a separately labeled sensitivity.

We allocate fractions $(0.5, 0.25, 0.1, 0.075, 0.075)$ of ϵ to statistical, algorithmic, logical, synthesis, and magic-state errors. For a spatial patch Ω and product formula with r steps,

$$|\widehat{D} - D_{\text{target}}| \le \epsilon_{\text{stat}} + |D_{\Omega,r} - D_\Omega| + |D_\Omega - D_{\text{target}}| + \epsilon_{\text{logical}} + \epsilon_{\text{synth}} + \epsilon_{\text{magic}}. \tag{6}$$

The truncation scenario splits the existing 0.25ϵ algorithmic reserve equally between Trotter and spatial bias. It does not add a new allowance. Mitigation assumes exact channel knowledge; a residual calibration bias would need its own allocation within Eq. (6). No unverified postselection bias reduction is credited in a resource curve.

4 Sampling, replication, and maximum lattice size

For inverse channels $\mathcal{E}_j^{-1} = \sum_a c_{ja}\mathcal{P}_{ja}$, define $\gamma_j = \sum_a |c_{ja}|$, $\Gamma = \prod_j \gamma_j$, and $\chi = \log \Gamma^2$. Centering the doublon observable as $D - I/2$ puts the PEC sample in $[-\Gamma/2, \Gamma/2]$. Hoeffding's bound then gives

$$n \ge \frac{\Gamma^2}{2\epsilon_{\text{stat}}^2} \log\frac{2}{\delta}, \qquad N_0 = \left\lceil \frac{\log(2/\delta)}{2\epsilon_{\text{stat}}^2} \right\rceil. \tag{7}$$

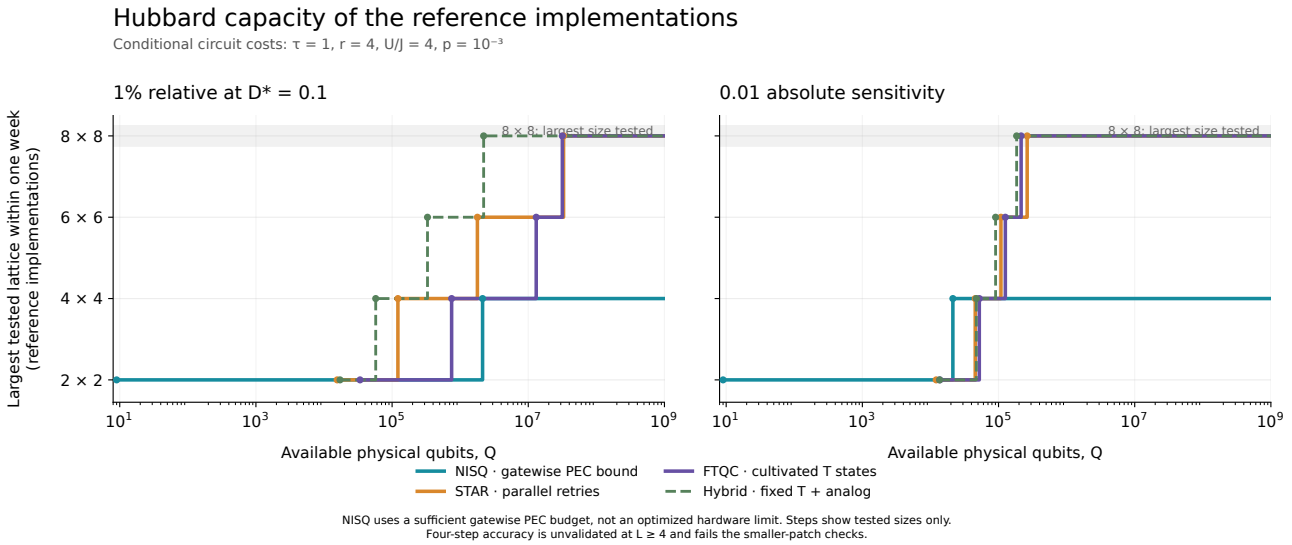


Figure 3: Largest tested Hubbard lattice within one week versus available physical qubits. Left: the accepted relative-precision reference. Right: the loose absolute-precision sensitivity. NISQ uses a sufficient gatewise-PEC sampling bound on the compiled observable cone at nominal idle noise; it does not represent optimized NISQ capacity. STAR chooses among one, two, four, and eight reference injection lanes; FTQC and hybrid use the packed cultivation supply scenario with zero additional handoff error. These are conditional $r = 4$ costs, not accuracy-qualified dynamics capacities. That step count fails the small square-patch checks, and $L \geq 4$ accuracy is uncalibrated. The NISQ compiler includes exact observable reductions that the encoded workloads have not yet received. Only $L = 2, 4, 6, 8$ is tested; the upper band marks that limit.

At strict reference precision, $N_0 = 7377759$. This is a sufficient common sampling convention, not a claim that every estimator needs this many shots.

Let q_a and t_a be the physical area and time of one complete sampler, and let a be whole-circuit acceptance. With $m = \lfloor Q/q_a \rfloor$ replicas,

$$T_{\text{answer}} \simeq \frac{N_0 e^{\chi} t_a}{am}, \qquad Q_{\min}(L) = q_a \max\left(1, \left\lceil \frac{N_0 e^{\chi} t_a}{aB} \right\rceil\right). \tag{8}$$

The code excludes a sampler whose single attempt exceeds B. Resource-state rejection is charged in delivery time, not counted again as whole-circuit rejection. Integer replicas are retained where they matter. Packing is an area-budget reference; a global square placement, calibration overhead, and control bandwidth are not yet modeled in the encoded curves. For PEC, $Q_{\min}$ is the minimum replica area that meets this sufficient sampling bound for the chosen implementation. It is not a lower bound on the qubits needed by another estimator, or even a tight cost for this one.

The principal output is now

$$L_{\max}^{\text{tested}}(Q) = \max\{L \in \mathcal{L}_{\text{tested}} : \min_{s} Q_{\min}(L, s) \leq Q\}, \tag{9}$$

where s labels allowed distances, supply configurations, and lane counts. This is a discrete upper envelope, not a smooth inverse fit. No feasible tested lattice is left blank, and reaching the largest tested size is marked as a survey boundary. We do not interpolate untested sizes or call that boundary a physical saturation. The companion output $\tau_{\max}(Q, L)$ follows by sweeping evolution time under the same contract; it has not yet been accuracy-qualified.

For a fixed circuit and estimator, Eq. (8) gives the exposure limit

$$\chi \lesssim \log \frac{amB}{N_0 t_a}. \tag{10}$$

Increasing hardware or runtime therefore buys logarithmic exposure, even though it can move a discrete lattice-size threshold appreciably. This is the mitigation wall of the specified strategy, not a universal NISQ law.

5 The common algorithm and its angle inventory

5.1 Termwise second order

In a fixed-number sector, the single-Z terms of the onsite interaction are a known phase. With $E = 2L(L-1)$ spatial bonds, the remaining Jordan–Wigner Hamiltonian has

$$K = 4E + N = 9N - 8L, \qquad \lambda = \sum_{\ell} |h_\ell| = 2E + \frac{|u|N}{4} = 5N - 4L. \tag{11}$$

There are four hopping Pauli strings per spatial bond, each with coefficient magnitude $1/2$, and one onsite ZZ per site, with magnitude $|u|/4$. Our convention is $U_P(\theta) = e^{-i\theta P}$; thus $R_z(\phi)$ has $\phi = 2\theta$, and a T gate has $|\theta| = \pi/8$ up to a global phase.

For a symmetric product over the K terms, choose onsite terms at both ends of the ordering and merge adjacent identical rotations at the center and between steps. Before further commuting or Clifford simplifications,

$$R_{\text{term}} = r(2K-2) + 1, \qquad \Theta_{\text{term}} = \tau\lambda. \tag{12}$$

The count follows from $r(2K-1)$ rotations after center merging and $r-1$ additional boundary merges. Every coefficient has total positive duration τ, proving the angle identity. Signed higher-order stages instead multiply the angle sum by their total absolute stage weight. The histogram, not merely Θ, is retained for channel evaluation.

5.2 Campbell plaquette second order with open boundaries

Campbell partitions hopping into disjoint pink and gold tiles and applies

$$S_2(\Delta) = e^{-i\Delta H_I/2} e^{-i\Delta H_p/2} e^{-i\Delta H_g} e^{-i\Delta H_p/2} e^{-i\Delta H_I/2}, \qquad \Delta = \tau/r. \tag{13}$$

We adapt the periodic construction to open boundaries. The pink partition has $P = L^2/4$ complete plaquettes. The shifted gold partition has $G = (L/2-1)^2$ complete plaquettes and $B = 2(L-2)$ isolated boundary bonds. Each bond belongs to exactly one partition, since $4(P+G) + B = E$.

A single-spin four-site hopping cycle has eigenvalues $\{2, 0, 0, -2\}$. Campbell's circuit implements its evolution with two variable-angle Pauli rotations and eight fixed T gates [7, Appendix E.1]. An isolated bond needs two commuting hopping Pauli rotations per spin. For $L \geq 4$, including both spin species and exact onsite boundary merging, this gives

$$R_{\text{plaq}} = N(r+1) + 4r(2P + G + B), \tag{14}$$

$$T_{\text{fixed}} = 16r(2P + G), \tag{15}$$

$$\Theta_{\text{dyn}} = \tau \left[\frac{|u|N}{4} + 4(P+G) + 2B \right]. \tag{16}$$

For $L = 2$ the gold partition is empty: the two pink half steps merge, so $2P$ becomes P in the count formulas. The angle formula is unchanged. At $L = 4$, $\tau = 1$, $r = 4$, these formulas give $(R_{\text{plaq}}, T_{\text{fixed}}, \Theta_{\text{dyn}}) = (288, 576, 44)$.

The compiled total angle includes $T_{\text{fixed}}\pi/8$. Consequently, the invariance of $\tau\lambda$ under refinement in Eq. (12) does not apply to all non-Clifford operations in Eq. (13). The fixed basis changes are an algorithm-dependent STAR exposure even when the variable rotations have no injection floor. Distilling those fixed angles is an immediate hybrid candidate. Hamming-weight phasing, cancellations between neighboring basis changes, and angle extraction remain further compiler optimizations. Campbell neglects fermionic-swap Clifford costs in his non-Clifford metric; our native nearest-neighbor compilation explicitly restores this motion.

6 Physical computation: gates, locality, and idle exposure

For a k-qubit depolarizing location with total nonidentity-Pauli probability p_k, set $s_k = 1 - 4^k p_k/(4^k - 1)$. The inverse has identity coefficient $1 + p_k/s_k$ and $4^k - 1$ coefficients $-p_k/[(4^k - 1)s_k]$, giving

$$\gamma_k = 1 + 2p_k/s_k, \qquad \chi_{\text{phys}} = \sum_{\ell \in \text{cone}} 2\log\gamma_{k_\ell} \simeq 4pG_2 + 4(p/10)G_1 + 4p_{\text{idle}}G_{\text{idle}} + \chi_{\text{SPAM}}. \tag{17}$$

Preparation and readout bit flips use $\gamma = 1/(1 - 2p)$. The continuous-time idle convention composes as

$$p_{\text{idle}}(\Delta t) = \frac{3}{4}\left[1 - (1 - 4p/3)^{\Delta t/(100\,\text{ns})}\right]. \tag{18}$$

Only channels in the compiled measurement's backward dependency cone enter the local PEC product. CPTP channels outside that cone cannot affect its expectation. This sufficient cone can still exceed the observable support needed at a specified tolerance.

6.1 Fermionic blocks and parallel motion

Two-mode hopping, density evolution, and fermionic Hadamards are extracted from the plaquette workload while preserving its noncommuting stage order. A number-conserving Gaussian gate has odd-sector matrix $G \in U(2)$, vacuum amplitude one, and double-occupation amplitude $\det G$. An Euler decomposition of G therefore compiles to local Z rotations and at most two native R_{ZZ} gates. In particular,

$$V(\theta) = e^{-i\theta(XX+YY)}, \qquad F_{\text{swap}} = (S \otimes S)V(\pi/4), \tag{19}$$

so a swap followed by hopping can fuse into the same pair of entanglers with a changed angle. Products of Gaussian gates on the same adjacent pair fuse through operations on disjoint wires. A commuting density rotation adds at most one entangler.

The compiler gathers independent plaquette groups into intervals of a live fermionic order using parallel odd-even transposition sorting. All swaps are adjacent physical fermionic swaps on a snake embedded in the square grid [11]. It then executes disjoint groups concurrently and tries six initial-order/local-route choices. A preparation permutation is free for the specified occupation product state, while the final wire map and Fock signs remain explicit. Using only snake edges leaves room for a better two-dimensional compiler. The trapped-ion experiment's all-to-all connectivity and free ordinary-SWAP relabeling are a different compilation assumption [9].

The first onsite stage is a scalar on the initial Fock state, and the last onsite stage commutes with the measured doublon. Removing them is exact for this task. A reverse traversal through even fermionic blocks then removes blocks on disjoint support. These reductions are available to every arm; the current encoded resource curves retain the original full logical workload.

$L = 4,\ \tau = 1,\ r = 4$	Entanglers	10 ns layers	Local χ
Greedy fused route with cone	3149	2187	35.297
Parallel route, full evolution	3204	1074	22.214
Parallel route with observable reductions	2648	964	20.017

The final row has $t_a = 9.84$ μs and exposure

$$\chi = 10.5927_{2q} + 4.1201_{1q} + 5.1681_{\text{idle}} + 0.1361_{\text{SPAM}}. \tag{20}$$

Its strict-precision threshold is $Q_{\min} \simeq 2.13 \times 10^6$, versus 12168 with zero idle on the same circuit. Idle has decreased substantially but remains material. The gain combines scheduling, placement, fusion, and observable algebra; zero idle is a sensitivity rather than a demonstrated alternate schedule.

6.2 What the million-qubit estimate means

The 4×4 circuit occupies 32 active qubits, packed into a 36-qubit square. At the strict precision, the reference assigns $\Gamma^2 = e^{20.017} \simeq 4.94 \times 10^8$ and a sufficient budget of $N_0\Gamma^2 \simeq 3.64 \times 10^{15}$ shots. At 9.84 μs per attempt, this requires 59239 copies to fit within one week, giving $Q_{\min} = 2132604$. The large area is entirely a sampling-throughput cost. For the same circuit, duration, and precision, removing idle noise gives $Q_{\min} = 12168$; retaining only two-qubit noise gives $Q_{\min} = 180$. These are noise ablations, not demonstrated improvements or alternative predictions for the declared device.

Experiments require an additional calibration of the NISQ model. Phasecraft and Quantinuum simulated a 7×4 lattice with 56 qubits, using training with fermionic linear optics (TFLO) and time-series smoothing [9]. The Willow study reached 6×6 with 72 fermionic-mode qubits, using TFLO, symmetry selection, and other application-specific processing [1]. Our gatewise PEC bound does not describe these estimators. Nor do those demonstrations establish our particular 0.001 absolute-error contract at every interacting data point. Their compilation is also a relevant benchmark: Willow's published 4×4 formula gives 1620 CZ gates at four steps, where our reference uses 2648 native R_{ZZ} gates. This comparison needs the respective gate implementations and schedules, not only a gate-count ratio. We therefore retain the current curve as a specified PEC reference; the resource frontier for experimentally informed physical estimators remains unestablished.

6.3 Exact circuit cones and physical truncation

For the original plaquette block ordering, the sufficient interior cones checked at $r = 1, 2, 4$ contain $(4r + 2)^2 = 36, 100, 324$ sites. The $r = 1$ circuits at $L = 8, 12, 16, 20$ all compile to 2001 native entanglers on 72 modes in that implementation. Ordered termwise sweeps can instead have global strict dependencies even at one step. Neither observation establishes the smallest Hamiltonian region needed at finite tolerance.

At fixed evolution time a local-observable truncation may eventually remove size dependence, whereas a time family such as $\tau \propto L$ probes growing physical reach. A truncation must receive a bias allowance in Eq. (6); it cannot be inferred solely from the strict circuit cone. Section 10 begins the finite-patch check and states what its outer boundary leaves unresolved.

7 Encoded protection and the logical-fault budget

Both encoded arms use the declared per-patch, per-cycle suppression reference

$$p_L(d) = A(p/p_{\text{th}})^{(d+1)/2}, \qquad A = 0.1, \quad p_{\text{th}} = 0.01. \tag{21}$$

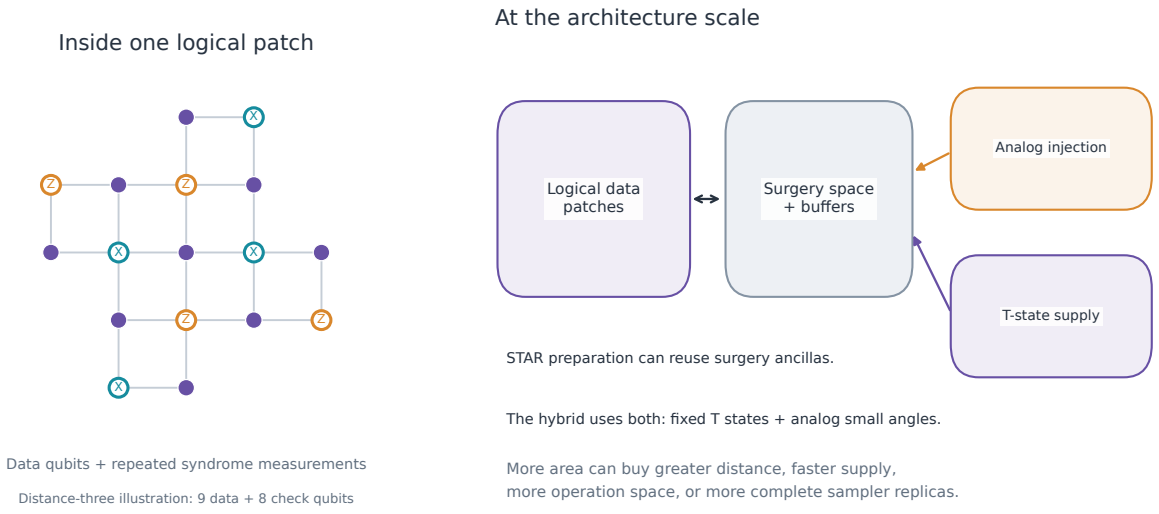


Figure 4: Physical and logical resource scales. Left: a distance-three rotated-code illustration with nine data and eight check qubits; links are nearest-neighbor data–check interactions. Right: logical data interacts through operation space with analog injection or T-state supply. STAR and FTQC use different supply paths; the hybrid uses both. The diagram shows functional roles: STAR preparation can occupy surgery ancillas rather than a separate supply region. This illustration is not a choice of distance or a calibrated surgery schedule for the resource curves.

The alternative $(A, p_{\mathrm{th}}) = (0.07, 0.0039)$ is motivated by the original STAR fit [2]. Its time convention and circuit noise must be reconciled before it becomes a calibration of this device. We use 240 ns per syndrome cycle and roughly $2d^2$ physical qubits per tile as reference accounting. Figure 4 distinguishes physical qubits, logical patches, and operation/supply space.

For V_L patch-cycles per attempt, $q_L \le V_L p_L(d)$ by a union bound. A $[0,1]$ unmitigated observable has fault bias at most q_L. A uniform branch-wise bound for centered PEC gives the sufficient worst-case Γq_L. Thus the reference chooses the smallest admissible odd d with

$$V_L p_L(d) \le \epsilon_L/\Gamma, \qquad d \gtrsim \frac{2\log(AV_L\Gamma/\epsilon_L)}{\log(p_{\mathrm{th}}/p)} - 1. \tag{22}$$

The implementation solves the discrete self-consistency problem because duration and volume also depend on d. Faults are budgeted per attempt; the full sampling campaign is not required to contain zero logical faults. Commuting, branch-independent fault channels can justify removing Γ locally. Section 8.1 checks this condition for a declared logical injection instrument, and Appendix A explains why independence alone is insufficient.

8 STAR: angle exposure, floors, and delivery

For independent residual dephasing q_j after completed rotations,

$$\Gamma^2 = \prod_j (1-2q_j)^{-2}, \qquad \chi_{\mathrm{STAR}} = -2\sum_j \log(1-2q_j) \simeq 4\sum_j q_j. \tag{23}$$

Original STAR uses the post-RUS reference $q_0 = 4p/15$ [2]. The small-angle reference uses $q_j = \alpha p|\theta_j| + f$, with $\alpha = 1.5$ in our angle convention and a declared floor $f = 10^{-5}$ [3, 8]. The floor is

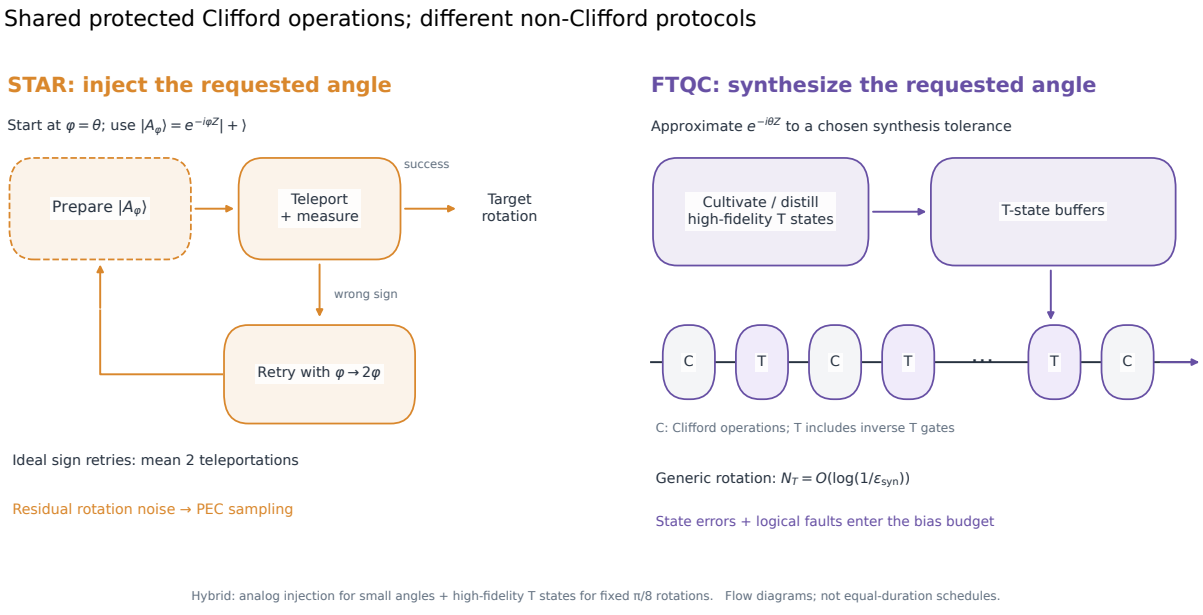


Figure 5: Two implementations of the same target rotation, in the convention $e^{-i\theta Z}$. STAR prepares an angle-specific resource state and uses teleportation with angle-doubling corrections [2, 3]. Ideal sign outcomes require two teleportations on average; this does not include rejected preparation attempts or their cost. FTQC consumes high-fidelity T states in a Clifford+T approximation, whose generic T count grows logarithmically with inverse synthesis tolerance [12]. The hybrid chooses between these paths by operation class. These are protocol flows, not schedules with equal durations or calibrated factory footprints.

a sensitivity motivated by injection and growth, not a measured error floor at this operating point. Fixed $\pi/8$ rotations retain q_0; we do not extrapolate the small-angle fit to them. Consequently,

$$\chi_{\mathrm{STAR}} \simeq 4\alpha p\Theta_{\mathrm{dyn}} + 4fR + \frac{16p}{15}T_{\mathrm{fixed}}. \tag{24}$$

The implementation evaluates the logarithmic product in Eq. (23); the physical channel models themselves remain approximations. Their three exposures have different responses to compilation and refinement. In particular, $fR \sim \alpha p\Theta_{\mathrm{dyn}}$ marks a floor/angle competition, while repeated basis changes can dominate separately.

The serial reference allocates $n_{\mathrm{data}} = \lceil 1.5(2N+6) \rceil$ tiles and six injection tiles. It charges 18 code clocks per rotation, with one clock equal to d syndrome cycles, plus startup, readout, and feedforward. This is a delivery model. A small-angle error law does not itself establish its acceptance, throughput, or footprint. The explicit retry refinement below adds both parallelism and its area cost.

8.1 Explicit retries, covariance, and parallelism

The logical injection instrument prepares $|A_\theta\rangle = e^{-i\theta Z}|+\rangle$, applies $\mathrm{CNOT}_{\mathrm{data}\to\mathrm{ancilla}}$, and measures the ancilla in Z. Its data Kraus operators are

$$K_0 = 2^{-1/2}e^{-i\theta Z}, \qquad K_1 = 2^{-1/2}e^{+i\theta Z}. \tag{25}$$

After outcome one, an injection of angle 2θ corrects a successful next branch: $K_0(2\theta)K_1(\theta) = K_0(\theta)/\sqrt{2}$. Repeating this gives the standard angle-doubling protocol underlying the STAR reference [2, 3]. CUDA-Q P0 now contains explicit preparation, CNOT, measurement, conditional retries,

and a returned exhaustion flag for caps $1, 2, 4, 8, 24$. This compilation does not supply a surface-code injection or patch-growth implementation.

For a Pauli fault on either wire immediately before or after CNOT, each conditional K remains diagonal or antidiagonal with equal-magnitude entries. Hence

$$K^\dagger K = I/2, \qquad KZ = \pm ZK. \tag{26}$$

The first identity makes the outcome probability independent of the data state; the second makes the conditional map commute with the channel $\mathcal{Z}(\rho) = Z\rho Z$. It therefore commutes with a Z-dephasing inverse, which is a linear combination of $\mathcal{I}$ and $\mathcal{Z}$. Mixtures of such faults and compositions over retries retain this local covariance under the stated branch-independent Pauli model. All sixteen two-qubit Paulis at both locations are checked independently.

This verifies the needed local commutation condition for that declared instrument. It does not establish branch-independent residual noise for physical state preparation, patch growth, measurement-dependent selection, or a mismatched channel inversion. The worst-case Γq_L bias bound therefore remains the default; q_L is a separately labeled covariance-based reference scenario. The physical STAR schedule still has to justify using it.

For b simultaneous ideal injections with geometric success probability $1/2$ and cap K, the mean batch retry time in units of one attempt is

$$\mathbb{E}[M_b] = \sum_{k=0}^{K-1} \Pr(M_b > k) = \sum_{k=0}^{K-1} \left[1 - (1 - 2^{-k})^b\right]. \tag{27}$$

It approaches 2 for one lane, but is approximately 3.50 for four lanes. Simply dividing the serial duration by four misses the slowest retry. The implemented policy has a barrier between batches, reserves six injection tiles per lane, and assumes nine clocks per attempt, recovering the previous 18-clock serial mean. It forms disjoint batches using full JW supports, not merely disjoint fermionic endpoints. An asynchronous policy could improve this conservative batch schedule.

With R rotations the ideal no-exhaustion probability is $(1 - 2^{-K})^R$. We choose K from a small allocated tail allowance and charge the acceptance penalty. At $L = 4$ and strict precision, two lanes minimize area at $Q = 121716$, versus 166698 for the earlier serial reference. Four and eight lanes reduce attempt time further but lose on area. At loose precision one lane still wins. These results constrain the particular logical mapping and resource model, not the best possible STAR architecture.

9 Full fault tolerance and the hybrid

For R synthesized rotations each with operator error ζ, telescoping unitaries bound the observable bias by $2R\zeta$. We set $\zeta = \epsilon_{\text{syn}}/(2R)$. The deterministic synthesis fit

$$T_R = \lceil 3.02 \log_2(1/\zeta) + 0.77 \rceil, \qquad N_T = T_{\text{fixed}} + RT_R, \tag{28}$$

has the expected logarithmic scaling [12]. At the compiled points, actual CUDA-Q GridSynth words replace this estimate. At $L = 4, \tau = 1, r = 4$, termwise Strang has 889 variable rotations, no fixed T gates, and 61178 synthesized T gates. Plaquettes have 288 variable rotations and 576 fixed T gates, for 19584 synthesized T gates: 66 per variable rotation. Independent matrix reconstruction checks the precision of every distinct emitted word. Whole-program synthesis at $L = 2, 4$ agrees with the class-weighted counts; $L = 6, 8$ use those counts before inter-gate optimization.

9.1 Conventional factory reference

Injected-state error p_{in} is separate from gate error p. The current conventional reference uses $46p/15$, with alternatives retained as sensitivities; this coefficient is not calibrated to our asymmetric $p_{1q} = p/10$ noise model. For a 15-to-1 stage [6],

$$a_1 = 1 - 15p_{\text{in}}, \quad q_1 = 35p_{\text{in}}^3, \quad q_2 = 35q_1^3. \tag{29}$$

One stage reserves twelve tiles, including its output, and takes $11/a_1$ mean code clocks. The deliberately serial two-stage reference reserves two eleven-tile blocks and fifteen storage tiles, so

$$n_F = 37, \qquad c_F = \frac{15(11/a_1) + 11}{1 - 15q_1}. \tag{30}$$

Both stages use the selected code distance, and protected factory faults enter V_L. Correlations, transport, and rejection-tail latency are not resolved by this leading-order mean-service model.

For F factory copies, the FT timing reference is

$$C_{\text{FT}} = 2 + \max(9N_T + 2R, \lceil N_T/F \rceil c_F), \tag{31}$$

$$t_a = C_{\text{FT}} d t_c + N_T t_{\text{ff}} + t_{\text{reset}} + t_{\text{meas}}, \tag{32}$$

$$q_a = 2d^2(n_{\text{data}} + F n_F). \tag{33}$$

Nine clocks per T is the compact-block consumption reference [6]; the $2R$ term is an extra Clifford-work placeholder. We optimize F over $\{1, 2, 4, 8, 16\}$ together with level and distance, subject to $N_T q_{1,2} \le \epsilon_{\text{magic}}$. Supply and consumption must be varied together: speeding consumption alone does not help a supply-limited circuit.

9.2 Why the hybrid is a separate candidate

The hybrid supplies the fixed T rotations as high-fidelity states while retaining analog small rotations. It removes the last term of Eq. (24), but pays factory area, supply time, and a magic-state budget amplified by the retained analog PEC reference. At fixed ideal shot budget, a proposed additional protection step improves the wall-clock cost when

$$\Delta \log t_a + \Delta\chi - \Delta \log a - \Delta \log m < 0. \tag{34}$$

This provides a marginal criterion for operation-dependent protection, observable-aware allocation, and mixed rotation delivery. It must include the lost replicas when protection consumes area. The earlier hybrid gain at $L = 4$ reversed under conventional two-level injection supply; that was an amortization effect of the serial reference, not a general disadvantage of the hybrid. Cultivation changes the comparison again, for both FT and hybrid, under the supply assumptions that follow.

9.3 Cultivation for both FT and hybrid supply

Gidney, Shutty, and Jones report cultivated-state error approximately 2×10^{-9} at $p = 10^{-3}$ and, in their Fig. 15, expected volume 6.6×10^4 qubit-rounds for the distance-five protocol with early rejection and sharing [4]. The output is in a grafted matchable code after escape to nominal distance 15; its behavior is not identical to an ordinary distance-15 surface-code patch. Holding, transfer, growth, and consumption require additional accounting.

Our declared packed-supply scenario rounds the expected volume to $V = 10^5$ qubit-cycles, takes footprint $q_f = 1024$, successful latency $\ell_f = 50$ cycles, and charges a $2d$-cycle handoff. These latter

choices are supply assumptions, not a compiled factory. The comparison also uses an unshared fixed-slot volume $V = 1024 \times 50/0.01 = 5.12 \times 10^6$. Mapping the paper's uniform physical noise and round definitions to this device's $p_{1q} = p/10$ and 240 ns cycle remains conditional.

For handoff error q_h, define

$$q_{\text{ready}} = 2 \times 10^{-9} + q_h, \qquad s_{\text{raw}} = \max(V/q_f, \ell_f) + 2d. \tag{35}$$

We sweep $q_h = 0, 10^{-9}, 10^{-8}$. Direct consumption is permitted only when

$$N_T q_{\text{ready}} \Gamma \le \epsilon_{\text{magic}}, \qquad \epsilon_{\text{magic}} = 0.075\epsilon, \tag{36}$$

with $\Gamma = 1$ for FT and the retained conservative analog-mitigation amplification for hybrid. Otherwise one additional 15-to-1 stage is available, with the same reference coefficients used previously [6]:

$$q_{\text{out}} = 35 q_{\text{ready}}^3, \qquad s_{\text{out}} = \frac{15 s_{\text{raw}} + 11d}{1 - 15 q_{\text{ready}}}. \tag{37}$$

The area includes the source footprint, one handoff buffer per source, and twelve further tiles if this stage is used. Surface-memory faults of the data, buffers, and final distillation stage enter the logical budget; the source preparation error already appears in Eq. (36). Supply and consumption run concurrently, so duration contains their maximum, and source copies and code distance are optimized under the week cutoff.

At strict precision and zero additional handoff error, direct unamplified consumption supports at most 37500 states. The compiled $L = 4$ FT count, 19584, fits; the $L = 6$ circuit requires the extra stage. Taking $q_h = 10^{-8}$ forces that stage already at $L = 4$. Cultivation therefore does not remove a distillation level independently of the task budget. In particular, the roughly 10^5 cutoff discussed around TopQAD Fig. 16 counts small-angle rotations at about 100 T states per rotation, not 10^5 T states [8].

$L = 4$, $r = 4$ reference	Q, $\epsilon = 10^{-3}$	Q, $\epsilon = 10^{-2}$
Serial STAR	166698	45486
FT, conventional serial factory	13791250	216890
FT, packed cultivation, $q_h = 0$	748656	52180
Hybrid, conventional serial factory	669714	54150
Hybrid, packed cultivation, $q_h = 0$	57472	47232

The conventional reference uses injected-state error $46p/15$. All rows remain conditional on $r = 4$. The loose-precision FT rows reuse the stricter synthesized words; resynthesis could improve them further. Cultivation changes the supply comparison appreciably, but neither these numbers nor their ranking should be promoted into a result for accurate dynamics.

10 Accuracy, truncation, and noisy observables

The calibration constructs H directly in fixed $(N_\uparrow, N_\downarrow)$ sectors. Sparse matrix exponentiation evaluates $e^{-i\tau H/J}$ without using the compiled gate decomposition. A separate evolution applies

$$U_r(\tau) = \left[e^{-i\Delta H_U/2} e^{-i\Delta H_{\text{pink}}/2} e^{-i\Delta H_{\text{gold}}} e^{-i\Delta H_{\text{pink}}/2} e^{-i\Delta H_U/2} \right]^r, \qquad \Delta = \tau/r, \tag{38}$$

where energies are in units of J. The largest patch is 3×3, with sector dimension $\binom{9}{4}\binom{9}{5} = 15876$. An independent eight-qubit dense Hamiltonian verifies the exact-sector observable; the original compiled plaquette circuit independently verifies the product formula on 2×2. The saved evidence contains 112 Trotter comparisons and 164 exact-curve points over four patches.

Patch, $\tau = 1$	Exact D	Error $r = 4$	Error $r = 16$	Error $r = 64$
2×2	0.1179790	0.0136397	0.00087075	0.00005447
3×3	0.1050849	0.0143939	0.00085350	0.00005317

For these square patches, $r = 64$ is the first tested power of two passing the strict split budget; $r = 16$ passes the loose split budget. These are not minimal-step claims. The asymptotic local error is consistent with r^{-2}, while coarse steps can be nonmonotonic. We also record a budget based on 1% of each actual patch signal, rather than silently replacing the provisional D_* in every resource curve.

The compiled $L = 2$ circuits at $r = 4, 16, 64$ contain respectively $164, 740, 3044$ native entanglers. The selected PEC estimator needs only one nine-qubit square copy at $r = 16$, but approximately 4.73×10^6 qubits at $r = 64$ to meet the week cutoff. Thus a common Trotter refinement can affect architecture costs very differently: it increases a physical noise exposure in an exponential, even when its algorithmic error reduction is shared. The corresponding $L = 4$ compilations demonstrate cost sensitivity only; their Trotter error has not been calibrated.

To start testing physical truncation, compare a five-site cross with its enclosing 3×3 square, keeping the same Néel pattern and central observable. Their exact differences are 4.75×10^{-4} at $\tau = 0.1$ and 7.93×10^{-2} at $\tau = 1$. Both exceed the strict truncation allowance. However,

$$|D_{\text{cross}} - D_{\text{target}}| \le |D_{\text{cross}} - D_{3\times3}| + |D_{3\times3} - D_{\text{target}}|, \tag{39}$$

and the second term is unknown. These finite-reference differences neither certify a bulk light cone nor bound the error of the larger patch. Larger patches, controlled local bounds, or both are still needed.

10.1 Postselection: measuring bias as well as acceptance

We propagate the full eight-qubit density matrix through the saved $L = 2$ native circuits. After each k-qubit primitive, the channel is

$$\mathcal{D}_{k,p}(\rho) = (1 - p)\rho + \frac{p}{4^k - 1} \sum_{P \neq I} P\rho P. \tag{40}$$

Primitive idle channels, initial preparation flips, and final readout flips are also included. A direct Pauli-mixture implementation independently checks the optimized channel calculation. For selection event A, the measured quantity is $D_A = \mathbb{E}[D\mathbf{1}_A]/\Pr(A)$; a large or small acceptance alone says nothing sufficient about its bias.

At $r = 4$, the unselected noise bias relative to the ideal Trotter circuit is 0.020371. Selecting the two spin parities reduces it to 0.005808 with acceptance 0.81958. Selecting exact $N_\uparrow = N_\downarrow = 2$ reduces it to 0.004212 with acceptance 0.80943. Its bias relative to exact dynamics, including the independently measured Trotter error, is 0.01785. At $r = 64$, the number-selected noise bias rises to 0.1070. Postselection helps here, but by itself does not meet the strict precision target.

The parity acceptance agrees with the independently propagated Pauli-character formula. Trace errors are below 1.4×10^{-12} and the calculated density matrices remain positive. These results apply to the stated eight-qubit circuits and noise channels. They do not prove failure of selection at $L = 4$ or of combinations with other mitigation methods. The larger trapped-ion experiment [9] remains a separate circuit, geometry, and noise comparison.

11 What determines the scaling

The exposure variables give a more transferable comparison than a single crossover. With local full-lattice compilation, $R = O(Nr)$ and $\Theta = O(N\tau)$. Physical PEC and original STAR therefore

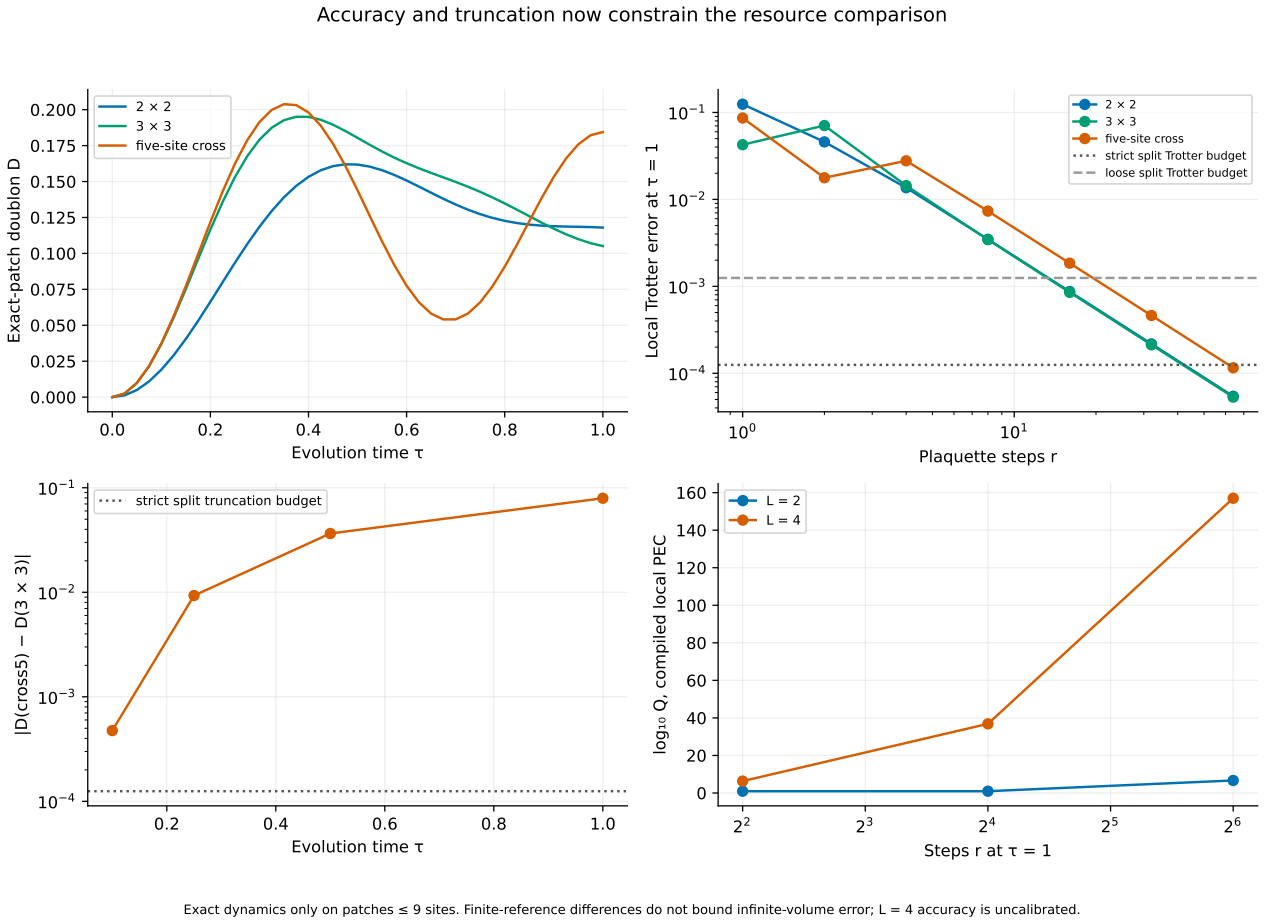


Figure 6: Exact patch dynamics, local product-formula errors, a nested spatial comparison, and the cost sensitivity to r. Each exact curve belongs to its own finite patch. The spatial difference is not an infinite-volume error certificate. The $L = 4$ cost sweep does not assume the small-patch error law holds at $L = 4$.

have reference exponents $O(pNr)$, while the small-angle term has exponent $O(pN\tau)$. An injection floor and the plaquette fixed-angle basis changes restore $O(r)$ growth of the STAR exponent. FT has $N_T = O(Nr \log(Nr/\epsilon))$ and logarithmic distance growth below threshold, but starts with a substantial protected footprint and supply overhead. These are laws of the specified models, subject to the locality and accuracy qualifications above.

Figure 3 expresses their current numerical consequences as a device-allocation question. At $Q = 10^6$ and strict reference precision, the largest tested conditional circuits are 2×2 for NISQ, 4×4 for STAR and FTQC, and 6×6 for the hybrid. These numbers use different implemented optimizations, conditional encoded supply, and the same unqualified $r = 4$ product formula. They are not final architecture rankings. At loose precision the encoded thresholds cluster much more closely, making delivery rate, distance, and integer replicas consequential.

Small-patch calibration explains why a common algorithmic correction does not cancel from the resource comparison: increasing r reduces a shared Trotter error but enters each architecture's cost differently. Likewise, physical truncation could reduce all three arms if its bias is controlled. The next scientific dependencies are a common accuracy-qualified workload, a controlled spatial truncation estimate, and physical injection/growth and handoff schedules with residual-channel validation.

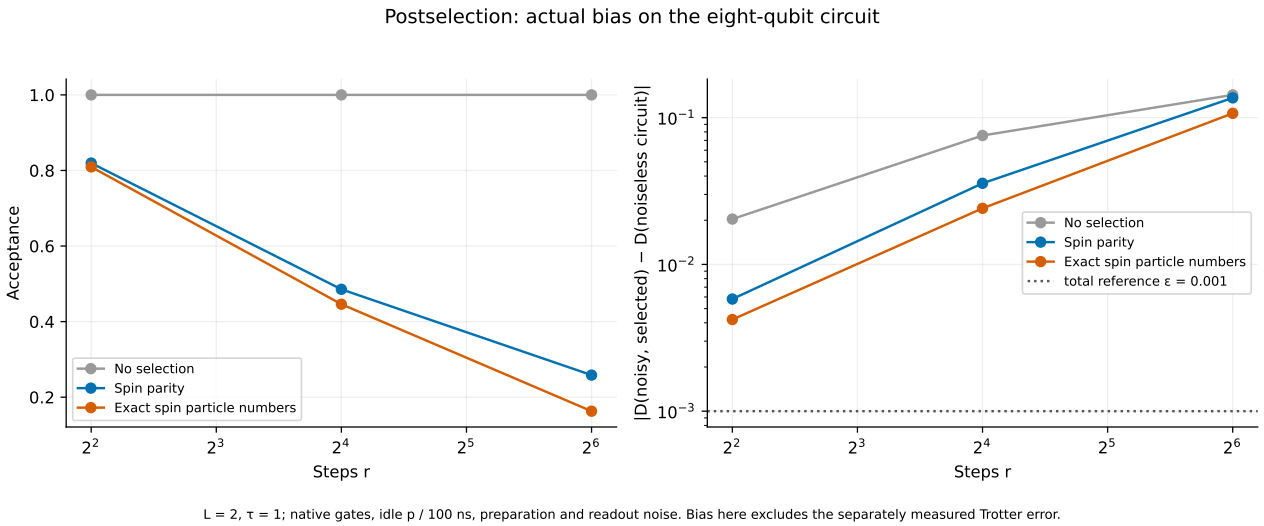


Figure 7: Exact acceptance and actual observable bias under the declared eight-qubit noise model. The right panel isolates noise bias from Trotter bias. No extrapolation from these curves to $L \geq 4$ is used.

Qubitization remains a separately identified later algorithm comparison.

A Logical-fault commutation and its limitation

A branch-independent fault channel $\mathcal{F}$ outside a complete PEC cancellation block, or commuting through its noise channel $\mathcal{E}$, satisfies $\mathcal{E}^{-1}\mathcal{F}\mathcal{E} = \mathcal{F}$. For a stochastic fault of probability q_L, this gives bias at most q_L for a $[0, 1]$ observable. Pauli channels commute locally, making this a useful condition. It does not follow merely from linearity or branch independence.

For a counterexample, take Bloch matrices $E = \mathrm{diag}(s, s, 1)$ and $F = (1 - q)I + qH$, where H is Hadamard conjugation. On $|0\rangle$, measuring $(I + X)/2$ after $E^{-1}FE$ gives bias $q/(2s) > q$ when $s < 1/2$. This is a stochastic non-Pauli fault inside the cancellation sandwich. The logical STAR instrument of Section 8.1 instead verifies local Z covariance under its stated Pauli fault model. A physical implementation must justify that model across injection, growth, retries, and any selection before the more favorable budget becomes its default.

B Compilation, checks, and reproducibility

The installed CPU CUDA-Q Logical environment compiles the common analog program, native physical circuits, and synthesized Clifford+T words. The physical routes carry mode permutations, explicit nearest-neighbor gates, ASAP schedules, counts, and hashes. The STAR retry program contains real logical preparation, measurement, and conditional angle doubling with an exhaustion flag. These P0 artifacts do not imply a physical STAR QEC schedule.

The installed minimal surface recipe lacks some needed gate gadgets. An explicit encoded macro adapter reaches physical IR, but the Hubbard backend timing is rejected. A reduced circuit with four fixed-angle rotations repeated four times, totaling 16 T states, triggers a nested schedule interval error. The 4-, 8-, and 32-T variants pass a loose bound, and separate two-resource template-call controls return their expected times. The root cause remains unresolved. None of these backend timings enters the resource curves; passing the loose controls is not a validation of physical encoded timing.

Tests independently check fermionic signs, product-formula identities, synthesis accuracy, exact-sector dynamics, primitive Pauli noise, STAR instrument algebra, and resource budgets. The capacity inversion additionally checks threshold inclusion, nonmonotone cost envelopes, and the untested region. These checks validate the specified calculations rather than the hardware assumptions.

The repository entry point is `README.md`. Source code is in `python/`, tests in `tests/`, assumptions in `configs/`, and this manuscript in `tex/main.tex` with a dedicated `tex/references.bib`. Current overview figures are in `figures/`; versioned scientific evidence stays in `results/`. Earlier complete notes are in `tex/archive/`. The original v3 manifest is preserved: historical source hashes are checked against their archived snapshots, while the relocated code replays the same saved numerical evidence. The presentation manifest records the new figures and their source datasets without relabeling older results as a new calculation.

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